

COUNTING HALLUCINATIONS IN DIFFUSION MODELS

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ABSTRACT

Diffusion probabilistic models (DPMs) have driven remarkable progress across generative tasks such as image and video synthesis, yet they persistently produce hallucinated samples that conflict with real-world knowledge—an implausible duplicate cup floating beside another, or a hand rendered with six fingers. Such failures are pervasive, but the absence of a feasible methodology for quantifying them systematically has impeded progress on the problem and obscured the design principles required for a next generation of generative models that respect factual constraints. We close this gap by isolating a well-scoped and rigorously measurable form of hallucination, which we term **counting hallucination**: the generation of an incorrect number of discrete instances or structured objects—such as a hand rendered with six fingers—despite the correct counts being fully determined by, and such deviations being entirely absent from, the training data. To render this phenomenon systematically studiable, we construct **CountHalluSet**, a dataset suite equipped with explicit, well-defined counting criteria and spanning a controlled spectrum of morphological complexity through its three constituent datasets: ToyShape, SimObject, and RealHand. On this foundation we establish a standardized, reproducible evaluation protocol for quantifying counting hallucinations and conduct a systematic analysis of how the principal sampling conditions of DPMs—solver type, ODE solver order, number of sampling steps, and initial noise—govern their incidence. This analysis surfaces a consistent decoupling between perceptual quality and factual correctness: numerical strategies that reliably sharpen perceptual quality, such as increasing the number of sampling steps, do not curb counting hallucinations when generating human hands. A dedicated correlation analysis corroborates this dissociation, establishing that widely used perceptual metrics—including the Fréchet Inception Distance (FID) and MUSIQ—track non-counting failures faithfully while failing to capture counting hallucinations, thereby exposing a systematic blind spot in prevailing generative evaluation. Motivated by these findings, we introduce joint-diffusion models, which denoise images jointly with explicit structural masks and measurably reduce counting hallucinations on RealHand, charting a concrete route toward factually consistent generation.

1 INTRODUCTION

Diffusion probabilistic models (DPMs) (Ho et al., 2020; Dhariwal & Nichol, 2021; Rombach et al., 2022), commonly referred to as diffusion models, have become the dominant paradigm in generative modeling, producing high-quality samples across a broad range of modalities that spans text-to-image generation (Ho & Salimans, 2021; Rombach et al., 2022), audio synthesis (Kong et al., 2021), video generation (Ho et al., 2022), 3D object generation (Luo & Hu, 2021; Vahdat et al., 2022), and protein structure prediction (Jumper et al., 2021; Abramson et al., 2024). These advances in perceptual fidelity, however, have not been matched by a comparable command of factual fidelity: diffusion models remain unable to reproduce faithfully the factual properties of their training data, and they routinely generate outputs that violate consistency with real-world knowledge. The most consequential manifestation of this deficiency is the generation of objects endowed with an incorrect number of spatially intricate components—supernumerary human fingers (Narasimhaswamy et al., 2024) or additional animal limbs (Borji, 2023) being canonical examples.

A growing body of work has sought to identify and characterize the common failure modes of text-guided diffusion models. Borji (2023) catalogued a range of qualitative shortcomings in text-to-image models; Liu et al. (2024) introduced an adversarial search procedure that surfaces undesirable

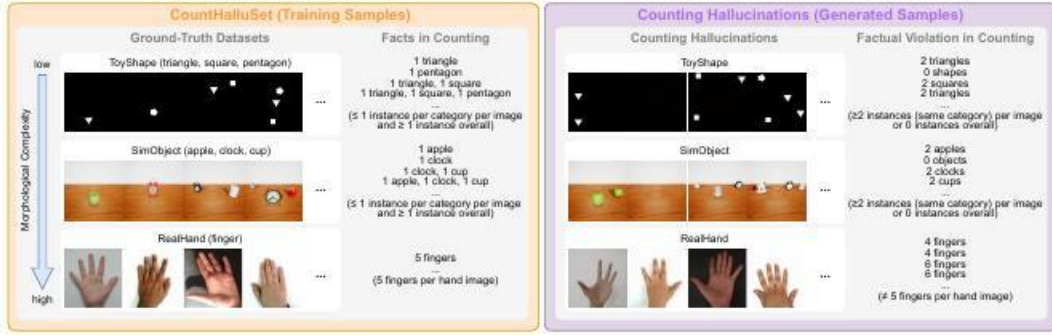


Figure 1: Comparison between the proposed **CountHalluSet** training data (left) and the corresponding **counting hallucinations** produced by models trained on it (right). Each dataset imposes an explicit counting criterion on its objects: in ToyShape and SimObject, every image contains at most one instance per category and at least one instance overall, whereas in RealHand every image contains exactly five fingers. Counting hallucinations are generated images that violate these criteria—for instance, two apples within a single image, or an image devoid of objects yet retaining a normal background—despite such configurations never appearing in the SimObject training data.

behaviors and failure cases in text-to-image generation; and Wu et al. (2023) developed an iterative method for discovering and mitigating spurious correlations. Collectively, these efforts establish that such failure modes are widespread; yet, being largely qualitative in character, they furnish no principled means of measuring how severe any given failure is—the very capability on which systematic progress depends.

More recently, *hallucination* has emerged as a distinct and especially insidious failure mode of diffusion models, attracting sustained attention from the research community (Aithal et al., 2024). Hallucinations differ fundamentally from the failure modes discussed above, and the distinction is what makes them so difficult to detect. Text-prompt-based failures can be assessed by measuring the alignment between input conditions and outputs (e.g., the CLIP score (Hessel et al., 2021) and the VQA score (Huang et al., 2025; Zarei et al., 2024)), and coarse failures such as severe distortions, blur, and artifacts are visually conspicuous; hallucinated samples, by contrast, typically appear visually plausible and arise independently of any conditioning signal, manifesting instead as subtle violations of factual consistency with the training data—incorrect object counts, geometries that look plausible yet are physically impossible, and analogous breaches of physical law. Precisely because these violations are fine-grained rather than conspicuous, the boundary between a valid sample and a hallucination constitutes a critical challenge for current AI systems, which detect such deviations only unreliably; even state-of-the-art large vision-language models such as GPT-4o fail at a task as elementary as counting the fingers in a clear hand image. Establishing a standardized, well-defined protocol for reliably quantifying hallucinated samples is therefore not merely useful but essential, for it furnishes the empirical foundation on which a next generation of generative models that enforce strict factual consistency must be built—a capability indispensable to their aspiration to serve as world models (Ha & Schmidhuber, 2018; Ding et al., 2024; Bruce et al., 2024).

Among the recurring categories of hallucinated content—counts, colors, spatial relationships, geometry, and physical realism—counting errors are uniquely amenable to systematic study, being at once readily identifiable and unambiguously quantifiable. We therefore ground our investigation in this specific and precisely measurable form of hallucination, which we term **counting hallucination**. In this setting the admissible number of objects is fixed in advance by the training data, so that a generated image qualifies as hallucinated exactly when its object counts deviate from these prescribed values. An image depicting a hand with six fingers, for instance, constitutes a counting hallucination because such a configuration never occurs in the RealHand dataset, as shown in the last row of Fig. 1. To interrogate this phenomenon across levels of visual difficulty, we construct the CountHalluSet suite, comprising three datasets that span a controlled spectrum of morphological complexity in the countable objects: ToyShape (triangle, square, pentagon), SimObject (mug, apple, clock), and RealHand (finger). Figure 1 presents representative examples from these datasets alongside their corresponding counting hallucinations.

The linchpin of our quantification protocol is the accurate identification and enumeration of objects, which we accomplish with dedicated counting models tailored to each dataset and validated against human annotation. Building on these datasets and counting models, we systematically characterize how the principal denoising conditions of diffusion models—the number of sampling steps, the initial noise, the order of the ODE solver, and the solver type—govern the counting hallucination rate. We then quantify, through a dedicated correlation analysis, the relationship between the counting hallucination rate and two widely used perceptual metrics, the Fréchet Inception Distance (FID) Heusel et al. (2017) and MUSIQ Ke et al. (2021).

Our principal contributions are as follows.

- **A formal definition and standardized protocol for counting hallucinations, together with the CountHalluSet suite.** We formalize counting hallucination as a measurable deviation from explicit, dataset-level counting criteria, and we operationalize this definition through a reproducible evaluation protocol backed by high-accuracy, human-validated counting models and the three-dataset CountHalluSet suite, which spans synthetic shapes, rendered objects, and real-world hands in a controlled progression of morphological complexity.
- **A systematic characterization of how denoising conditions govern counting hallucinations.** We show that the numerical strategies conventionally relied upon to improve diffusion sampling—increasing the number of sampling steps from 25 to 100, for example—fail to mitigate counting hallucinations when generating human hands, even where the same strategies demonstrably reduce them on simpler synthetic datasets. Perceptual improvement is thus decoupled from factual correctness of object counts, a dissociation that grows sharper as morphological complexity increases.
- **Evidence that perceptual metrics do not capture counting hallucinations.** Across datasets and solver configurations, the counting hallucination rate correlates only weakly and inconsistently with perceptual metrics such as FID and MUSIQ, whereas the non-counting failure rate correlates strongly and robustly with them—exposing a systematic blind spot at the heart of prevailing generative evaluation.
- **Joint-diffusion models, a training paradigm for factual generation.** Motivated by our finding that morphological simplicity aids counting fidelity, we propose joint-diffusion models (JDM), which denoise images jointly with explicit structural masks and, on RealHand, reduce both counting hallucinations and non-counting failures relative to a vanilla latent diffusion baseline, albeit at some cost in FID. We present JDM as an initial but concrete demonstration that CountHalluSet supports not only diagnosis but also method development.

2 RELATED WORK

2.1 HALLUCINATIONS IN DIFFUSION MODELS

Prompted by the intense scrutiny of hallucination in large language models (LLMs) (Kadavath et al., 2022; Ji et al., 2023; Huang et al., 2023), the study of hallucinations in unconditional diffusion models has attracted growing interest. Most notably, Aithal et al. (2024) advanced a qualitative hypothesis attributing hallucinated samples to interpolation between distinct data modes. Valuable as this account of potential causes is, it remains qualitative; a systematic *quantitative* analysis of hallucinations in diffusion models has to date been absent—precisely the gap this work sets out to close.

Accordingly, we develop the machinery for accurately identifying and quantifying hallucinations in diffusion-generated outputs. Such a quantitative account serves two ends at once: it deepens our understanding of the undesirable behaviors these models exhibit, and it makes explicit how the magnitude of denoising error—as modulated by the number of sampling steps and the order of the ODE solver—bears on the incidence of hallucination.

2.2 QUANTITATIVE METRICS IN IMAGE GENERATION

The Visual Question Answering (VQA) score (Huang et al., 2025; Zarei et al., 2024) and the CLIP score (Hessel et al., 2021) are the two metrics most widely used to evaluate text-guided image

generation. Evaluation of this kind, however, inherits a fundamental limitation from its reliance on language-based models: it is poorly suited to assessing hallucinated images. Two independent lines of evidence render this limitation concrete. First, there exists a statistical lower bound on the propensity of pretrained language models to hallucinate certain classes of facts, which in turn upper-bounds the detection accuracy attainable by language models alone (Kalai & Vempala, 2024). Second, a persistent performance gap separates human annotators from LLMs in classifying hallucinated textual content (Lin et al., 2022; Li et al., 2023).

A complementary family of quantitative metrics—FID (Heusel et al., 2017), IS (Salimans et al., 2016), Precision & Recall (Sajjadi et al., 2018), and MUSIQ (Ke et al., 2021)—is routinely employed in image generation, each of them computed from feature embeddings extracted by a pretrained Inception-v3 model (Szegedy et al., 2016). Whether these perceptual metrics can capture the severity of hallucinations in generated data has nonetheless remained an open question. We resolve this question directly, and in the negative, from the vantage point of counting hallucinations.

3 PRELIMINARY: DIFFUSION PROBABILISTIC MODELS

3.1 FORWARD (DIFFUSION) PROCESS

Forward Process. Given observed data $\mathbf{x}_0 \sim q_0(\mathbf{x}_0)$, Diffusion Probabilistic Models (DPMs) (Sohl-Dickstein et al., 2015; Ho et al., 2020) define a *forward process* (equivalently, a *diffusion process*) that progressively corrupts the data by injecting Gaussian noise over T discrete timesteps. At each timestep t , the marginal distribution of the noisy sample $\mathbf{x}_t$ conditioned on the original data $\mathbf{x}_0$ takes the form

$$q_{0:t}(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t | \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad \alpha_t := 1 - \beta_t, \quad \bar{\alpha}_t := \prod_{j=1}^t \alpha_j, \quad (1)$$

where $t \in \{0, 1, \dots, T\}$ and $\{\beta_t\}_{t=1}^T$ is a predefined, t -dependent noise schedule, most commonly instantiated as a linear (Ho et al., 2020) or cosine (Nichol & Dhariwal, 2021) schedule. The factor $\bar{\alpha}_t$ governs the signal-to-noise ratio at step t , interpolating monotonically from $\bar{\alpha}_0 = 1$ (no noise) to $\bar{\alpha}_T \approx 0$ (nearly pure noise). The reparameterization trick renders Eq. (1) equivalent to

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (2)$$

As T grows sufficiently large, the marginal distribution $q_{0:T}(\mathbf{x}_T | \mathbf{x}_0)$ converges to a standard Gaussian $\mathcal{N}(\mathbf{0}, \mathbf{I})$, which supplies the tractable prior from which sampling is initialized. In widely used DPMs (Ho et al., 2020; Nichol & Dhariwal, 2021), T is conventionally set to 1000.

3.2 REVERSE (DENOISING) PROCESS.

Training Objective. The *reverse process* (equivalently, the *denoising process*) of DPMs iteratively reconstructs the observed data from the diffused sample $\mathbf{x}_T$. Concretely, a neural network $\epsilon_\theta(\mathbf{x}_t, t)$ —the *noise prediction model*—is trained to recover the injected noise $\boldsymbol{\epsilon}$ from the noisy sample $\mathbf{x}_t$. Its parameters θ are obtained by minimizing the objective (Ho et al., 2020; Song et al., 2022)

$$\min_{\theta} \mathbb{E}_{t, \mathbf{x}_0, \boldsymbol{\epsilon}} [\pi(t) \|\epsilon_\theta(\mathbf{x}_t, t) - \boldsymbol{\epsilon}\|_2^2], \quad (3)$$

where $\pi(t) > 0$ is a per-timestep weighting function that balances the contribution of each noise level to the total loss.

Ancestral Sampling. Given a noisy sample $\mathbf{x}_T$, one draws samples from the data distribution $q_0(\mathbf{x}_0)$ by reversing the diffusion process over the same number of steps, a procedure known as *ancestral sampling*:

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} | \frac{1}{\sqrt{1 - \beta_t}} (\mathbf{x}_t + \beta_t \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)), \beta_t \mathbf{I}), \quad (4)$$

in which the sole unknown term $\nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ is the *score function* (Song et al., 2022).

In practice, DPMs approximate the scaled score function $-\sqrt{(1 - \bar{\alpha}_t)} \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ by means of the noise prediction model, whereupon Eq. (3) admits the equivalent formulation

$$\mathbf{x}_{t-1} = \boldsymbol{\mu}_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \boldsymbol{\epsilon}, \quad \boldsymbol{\mu}_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{1 - \beta_t}} \mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \beta_t}} \frac{1}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t). \quad (5)$$

ODE-Based Accelerated Sampling. Whereas ancestral sampling in Eq. (4) incorporates a stochastic term (Kloeden et al., 1992), it admits a deterministic counterpart expressible as an ordinary differential equation (ODE)—the *probability flow ODE* (Song et al., 2022):

$$\frac{d\mathbf{x}(\tau)}{d\tau} = f(\tau)\mathbf{x}(\tau) + h(\mathbf{x}(\tau), \tau), \quad f(\tau) = -\frac{1}{2}\beta(\tau), \quad h(\mathbf{x}(\tau), \tau) = -\frac{1}{2}\beta(\tau)\nabla_{\mathbf{x}(\tau)} \log q_\tau(\mathbf{x}(\tau)), \quad (6)$$

where $\tau \in [T, 0]$. Let $\mathbf{x}(\tau)$ denote the ideal trajectory that satisfies this ODE under the true score function $h(\mathbf{x}(\tau), \tau)$, initialized from ideal diffused data $\mathbf{x}(T)$ drawn from the true diffused distribution $q_T(\mathbf{x}(T))$. Discretizing time as $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$ with $\tau_N = 0$, and applying the variation-of-constants formula to the exact solver over a single step, yields

$$\mathbf{x}(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k)\mathbf{x}(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u)h(\mathbf{x}(u), u)du \quad (7)$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v)dv}$ is the state transition operator for the linear part $\frac{dz}{d\tau} = f(\tau)z$, and $\Delta t > 0$ is the step size. Because the ODE formulation admits step sizes substantially larger than those of ancestral sampling, it accelerates generation while requiring only $T/\Delta t$ total sampling steps. Since the linear part $\mathcal{G}(t_2, t_1)$ possesses a closed-form solution, most accelerated ODE-based samplers (Song et al., 2021; Liu et al., 2022; Zhang & Chen, 2023; Lu et al., 2022; 2023) direct their numerical effort—by way of higher solver orders and finer step sizes—at approximating the nonlinear integral term involving h as accurately as possible. The corresponding theoretical analysis, which traces how these approximation choices bound the accumulated sampling error, is developed in Appendix A.1 through A.3.

4 FORMAL DEFINITION, DATASETS, AND EVALUATION PROTOCOL FOR COUNTING HALLUCINATIONS

This section lays out the methodology and datasets that underpin our quantification of counting hallucinations. We first formalize the notion of counting criteria and define counting hallucinations precisely as deviations from them; we then introduce the three datasets that constitute the CountHalluSet suite. For each dataset we specify its counting criteria and the counting hallucinations they induce, and we detail the standardized protocol by which those hallucinations are measured.

4.1 COUNTING HALLUCINATIONS: DEVIATIONS FROM THE COUNTING CRITERIA

We now give a general definition of the counting criteria that a training dataset imposes, and we characterize counting hallucinations precisely as violations of these predefined constraints.

Counting criteria in a training dataset. Let $\mathcal{D}_{\text{ref}}$ denote a training dataset. For any reference sample $\mathbf{x} \in \mathcal{D}_{\text{ref}}$, the number of objects of category c , denoted $N_c(\mathbf{x})$, is constrained to lie within a predefined admissible set $\mathcal{S}_c \subseteq \mathbb{Z}_{\geq 0}$ (e.g., $\{0, 1\}$ or $\{1, 2, 3\}$); that is, $N_c(\mathbf{x}) \in \mathcal{S}_c$. We additionally require every valid reference sample to contain at least one object, $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$, where C denotes the set of admissible categories. Together these two conditions constitute the counting criteria of $\mathcal{D}_{\text{ref}}$, and by construction every training sample satisfies them.

Counting hallucinations as deviations from the counting criteria. A generated sample $\hat{\mathbf{x}}$ is said to exhibit a *counting hallucination* with respect to $\mathcal{D}_{\text{ref}}$ if, and only if, the following two conditions hold jointly:

1. **Sufficient visual quality for counting.** The image $\hat{\mathbf{x}}$ must be of sufficient visual quality that its background and every present object are clearly discernible. This condition cleanly separates counting hallucinations from other prominent failure modes, such as severe degradations, that would render counting itself ill-defined. We encode it through a binary *counting-ready indicator* $\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \in \{0, 1\}$, in which 1 signifies that the image is of adequate quality for counting and 0 signifies otherwise. The indicator may be instantiated by human annotation, by a classifier calibrated against human annotations, or by any other automated criterion, affording the protocol flexibility across datasets of differing difficulty.
2. **Violation of counting facts.** A violation of counting facts occurs whenever some category c satisfies $N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$, or the image contains no objects whatsoever ($\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$), the latter case violating the requirement of at least one object per sample. The requisite object counts are obtained by applying a dedicated counting model to each generated image.

A counting hallucination (CH) is thus declared precisely when the image is counting-ready ($\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) = 1$) and its predicted counts violate the counting criteria ($\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$ or $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$). Formally,

$$\mathbb{I}_{\text{CH}}(\hat{\mathbf{x}}) = \mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \wedge \left(\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c \vee \sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0 \right).$$

4.2 DATASETS, COUNTING HALLUCINATIONS AND EVALUATION PROTOCOL

The CountHalluSet suite comprises three datasets purpose-built to probe counting hallucinations across a controlled progression of morphological complexity: ToyShape, SimObject, and RealHand. Representative examples from each dataset, together with their corresponding counting hallucinations, are shown in Fig. A1 and A2, while complete statistical summaries appear in Appendix A.4.

4.2.1 TOYSHAPE

Dataset description. The ToyShape dataset consists of 30,000 images populated by three geometric shapes—*triangle*, *square*, and *pentagon*. Every shape is rendered in white, occupies an equal area of 120 pixels, and is placed so that no two shapes overlap within an image, eliminating occlusion as a confound. The counting criteria stipulate that each image contain at most one instance of each shape category and at least one shape overall, as illustrated in the top-left of Fig. 1. Formally, $C = \{\text{triangle}, \text{square}, \text{pentagon}\}$, and for each class $c \in C$ we set $N_c(\mathbf{x}) \in \mathcal{S}_c = \{0, 1\}$ subject to the constraint $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Counting hallucinations. Under the criteria defined above, any image generated by a model trained on ToyShape that contains two or more shapes of the same category constitutes a counting hallucination—for example, the image with two pentagons in the top-right of Fig. 1. Empty images, displaying only the black background and no shapes, likewise qualify as counting hallucinations, since they violate the at-least-one-object requirement. Additional examples appear in Appendix A.5.

Counting model used in ToyShape. We identify counting errors with a dedicated counting model and dispense with the counting-ready indicator, as the dataset’s simplicity renders severely degraded images negligibly rare. Specifically, we fine-tune a ResNet-50 (He et al., 2016) on over 400k ToyShape images in which each category appears 0–3 times per sample (i.e., $\mathcal{S}_c = \{0, 1, 2, 3\}$); this deliberately broad label range spans the full space of plausible configurations—encompassing both counting-valid samples and counting hallucinations—and thereby underwrites accurate quantification. The resulting model attains over 99.9% counting accuracy on generated data, a level of reliability that justifies treating its predictions as ground truth for ToyShape.

Quantification of counting hallucinations. Given a diffusion model trained on ToyShape, we generate a sample set matched in size to the training set (30,000 images), apply the counting model to predict the object counts in each generated image, and flag as a counting hallucination every image that violates the counting criteria, as illustrated in Fig. 2.

4.2.2 SIMOBJECT

Dataset description. The SimObject dataset comprises 30,000 rendered images of everyday objects drawn from three categories—*mug*, *apple*, and *clock*—each realized in 10 distinct intra-class variations. Objects are placed at random on a wooden tabletop under fixed lighting and with minimal mutual occlusion, as shown in the middle-left of Fig. 1. The images are synthesized in Unreal Engine 5 (Epic Games, 2024), whose photorealistic rendering faithfully simulates object geometry, material properties, lighting, and reflections; SimObject thereby introduces the appearance complexity absent from ToyShape while preserving an equally unambiguous counting structure, and so serves as the intermediate rung of the suite’s complexity ladder. The counting criteria parallel those of ToyShape (see Sec. 4.2.1): formally, $C = \{\text{mug}, \text{apple}, \text{clock}\}$, and for each class $c \in C$ we set $N_c(\mathbf{x}) \in \mathcal{S}_c = \{0, 1\}$ subject to $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Quantification of counting hallucinations. Because its counting criteria coincide with those of ToyShape, SimObject inherits the same definition of counting hallucinations, the same counting-model construction, and the same quantification protocol, as shown in Fig. 2.

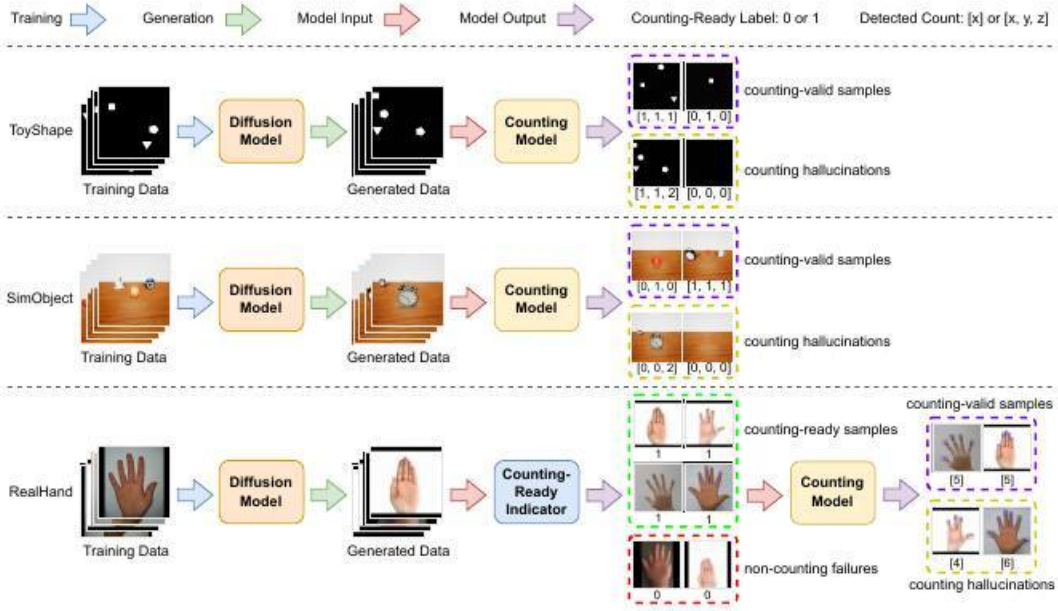


Figure 2: The evaluation procedure of the CountHalluSet suite. For ToyShape and SimObject, generated images are counted directly. For RealHand, a counting-ready indicator—a binary classifier aligned with human judgments—is first applied to separate counting hallucinations from non-counting failures such as severely deformed fingers, after which counting proceeds only on the retained, counting-ready images. Count labels correspond to ToyShape: *triangle, square, pentagon*; SimObject: *mug, apple, clock*; RealHand: *finger*.

4.2.3 REALHAND

Dataset description. To probe counting hallucinations in a demanding real-world setting, we construct the RealHand dataset of 5,050 human hand images aggregated from the 11k Hands dataset (Afifi, 2019), Kaggle¹, and Roboflow², with blurred or artifact-laden images removed during curation. Each image depicts a five-fingered human hand, spanning both dorsal (back) and palmar (front) views under varied lighting and backgrounds (white surfaces, blackboards, walls), as shown in the lower-left of Fig. 1. The counting criteria stipulate exactly five visible fingers per image: $C = \{finger\}$ with $N_c(x) \in S_c = 5$ for each $c \in C$, so that $\sum_{c \in C} N_c(x) = 5$. This exactly-five criterion is a deliberately controlled operationalization: it fixes a single, unambiguous count that the curated distribution satisfies, at the cost of conflating anatomical correctness with viewpoint and finger visibility, and it is therefore best read as a tractable proxy for counting fidelity on hands rather than as a universal definition applicable to arbitrary real-hand distributions.

Counting hallucinations and non-counting failures. The generation of realistic hand images is intrinsically prone to failures that extend beyond counting hallucinations (e.g., extra or missing fingers) to encompass blurriness, severely deformed fingers, and visual artifacts. We therefore introduce a counting-ready indicator—a binary classifier—to isolate these non-counting failures prior to counting, thereby preventing them from contaminating the counting-hallucination measurement. Examples of counting hallucinations and non-counting failures are provided in Appendix A.7.

Counting-ready indicator and counting model used in RealHand. To separate counting hallucinations from the failure modes above, we fine-tune a MaxViT (Tu et al., 2022) model on 2.5k annotated generated images to serve as the counting-ready indicator aligned with human judgments, and we fine-tune a YOLO-12 (Tian et al., 2025) detector on 2k fingertip-annotated generated images to perform the counting. Evaluated by three annotators on 500 randomly sampled images, the indicator and the counting model achieve over 96% and 99% accuracy, respectively—a level of agreement with human judgment that supports their use as ground-truth surrogates in the RealHand protocol. We nonetheless emphasize that the measured rates on RealHand are conditional on this detection

¹<https://www.kaggle.com/>

²<https://universe.roboflow.com/>

pipeline and its operating thresholds; residual detector failures on borderline images, together with the sensitivity we characterize in Appendix A.6, bound the precision attainable in this real-world regime. Representative YOLO detections appear in Appendix A.6.

Quantification of counting hallucinations. Unlike ToyShape and SimObject, whose simplicity and near-absence of severe failures permit direct evaluation of every generated image, RealHand requires an additional filtering stage. We first apply the counting-ready indicator to retain only hand images of sufficient visual quality, and only then invoke the counting model to predict the number of fingers, as shown in Figure 2.

5 EXPERIMENTS

5.1 IMPLEMENTATION DETAILS

Training of Diffusion Models. For ToyShape we train a DDPM (Ho et al., 2020) from scratch, whereas for SimObject and RealHand we fine-tune latent diffusion models (LDMs) (Rombach et al., 2022) pretrained on the CelebA-HQ dataset (Karras et al., 2018). The number of timesteps is fixed to $T = 1,000$ under a standard linear noise schedule, and images are resized to 128×128 for the DDPM and 256×256 for the LDMs. All models are optimized with Adam (Kingma, 2015) at a learning rate of 0.0001 and an effective batch size of 256, run for 150k, 300k, and 80k steps on ToyShape, SimObject, and RealHand, respectively.

Inference of Diffusion Models. For ancestral sampling (DDPM) we use the full 1,000 steps. For the ODE-based solvers we examine both first- and second-order methods: DPM-Solver-1—equivalent to DDIM (Song et al., 2021)—as the first-order solver, and DPM-Solver-2 (Lu et al., 2022) as the second-order solver, each evaluated under the three widely used step budgets of 25, 50, and 100. For RealHand, the YOLO-based counting model operates with a confidence threshold of 0.3 and an IoU threshold of 0.1, a single configuration held fixed across all RealHand evaluations to ensure comparability. Unless stated otherwise, the number of generated samples matches the training-set size in every evaluation, and all reported results are averaged over three random seeds, so that the observed effects reflect systematic behavior rather than sampling variance.

5.2 QUANTIFYING COUNTING HALLUCINATIONS UNDER DIFFERENT DENOISING CONDITIONS

Table 1 reports the counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR) across the three datasets—ToyShape, SimObject, and RealHand—as functions of the denoising conditions: solver type, ODE solver order, number of sampling steps, and initial noise. The results support five consistent and mutually reinforcing observations, which together characterize how sampling configuration governs counting fidelity in this DDPM/LDM setting.

(1) Increasing sampling steps reduces counting hallucinations on synthetic datasets but exacerbates them on RealHand. Under the common ODE solver budgets of 25, 50, and 100 steps, adding sampling steps monotonically lowers the CHR on ToyShape and SimObject yet raises it on RealHand, the sole exception being DPM-Solver-2 at 100 steps (Table 1). This dissociation admits a coherent reading: the simpler, more structured distributions of synthetic data genuinely benefit from finer solver granularity, whereas for the far more complex distribution of real hands the additional steps instead fit local inconsistencies and thereby amplify hallucinations. The standard “more steps is better” heuristic thus does not extend to counting fidelity in the real-world regime.

(2) Higher-order ODE solvers suppress non-counting failures but aggravate counting hallucinations on RealHand at standard step budgets. Comparing DPM-Solver-1 and DPM-Solver-2 on RealHand at the standard budgets of 25 and 50 steps exposes a consistent trade-off (Table 1): the higher-order solver sharply reduces non-counting failures while simultaneously raising the counting hallucination rate, and the same pattern recurs as the step count grows for either solver. The implication is clear: allocating more computation to “better” sampling reliably improves perceptual quality, as registered by a lower NCFR, but does not, of itself, enforce the factual correctness of finger counts.

(3) DDPM decisively outperforms the ODE solvers in suppressing both counting hallucinations and non-counting failures. Across every comparison, ancestral sampling (DDPM) attains the lowest

Table 1: Counting hallucination rate (CHR; the number of counting hallucinations divided by the total number of generated samples) across solver configurations and initial-noise settings on the three datasets ToyShape, SimObject, and RealHand. For RealHand we additionally report the non-counting failure rate (NCFR)—the fraction of failures caught by the counting-ready indicator—and the total failure rate (TFR), defined as the sum of CHR and NCFR. *Diffused* and *Normal* denote the ground-truth initial noise and standard Gaussian noise, respectively. Lower values are better throughout. The central comparison is the divergence, as morphological complexity rises from ToyShape to RealHand, between how added sampling steps affect CHR and how they affect NCFR.

Solver Name	Sampling Steps	CHR (ToyShape)		CHR (SimObject)		CHR (RealHand)		NCFR (RealHand)		TFR (RealHand)	
		<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>
DPM-Solver-1 (Song et al., 2021)	25	2.43	3.47	9.27	9.95	12.71	12.95	16.18	18.06	28.90	31.01
	50	1.83	2.79	8.53	9.27	13.82	13.85	11.32	12.51	25.14	26.36
	100	1.56	2.32	8.04	8.90	14.29	14.55	9.54	10.63	23.83	25.19
DPM-Solver-2 (Lu et al., 2022)	25	2.45	3.81	8.16	8.19	14.23	14.48	8.37	9.33	22.60	23.82
	50	1.76	2.86	7.90	7.95	16.15	15.99	6.38	7.22	22.53	23.21
	100	1.41	2.24	7.89	7.89	15.06	15.43	7.82	8.94	22.88	24.38
DDPM (Ho et al., 2020)	1000	0.63	0.64	4.98	4.99	10.82	10.75	2.07	2.39	12.88	13.14

CHR, NCFR, and TFR simultaneously, establishing it within this experimental setting as a practical lower bound on the achievable failure rates and, correspondingly, as the most effective sampling strategy for minimizing hallucinations.

(4) **A better-informed initial noise reduces both counting hallucinations and non-counting failures.** As Table 1 shows, “diffused” noise—the ground-truth initial noise—consistently yields lower CHR, NCFR, and TFR than “normal” standard Gaussian noise. This gain follows naturally from the closer alignment of diffused noise with the initialization distribution encountered during training, and it establishes that a more informed initialization mitigates not only counting hallucinations but non-counting failures as well.

(5) **Greater morphological complexity of the countable objects increases the incidence of counting hallucinations.** Reading the CHR across ToyShape, SimObject, and RealHand reveals a clear monotonic trend: hallucination rates rise systematically with the morphological complexity of the target objects. The fine-grained detail and intricate spatial relationships of morphologically complex objects thus pose a genuinely harder counting problem for diffusion models, precipitating more frequent additions and omissions. This monotonic dependence is the empirical lever we exploit in Sec. 5.5 to motivate joint-diffusion models.

5.3 CORRELATION ANALYSIS BETWEEN COUNTING HALLUCINATION RATE, NON-COUNTING FAILURE RATE, AND PERCEPTUAL METRICS

Since counting hallucination is a form of factualness hallucination, a central question follows naturally: *does an improved perceptual metric, such as a lower FID, necessarily entail fewer counting hallucinations by reflecting a smaller distribution gap between generated and training data?* Our evidence answers this question in the negative: it does **not**.

To establish this, we conduct a correlation analysis relating both CHR and NCFR to perceptual metrics. We study two complementary and widely used measures—the distribution-level FID (Heusel et al., 2017) and the instance-level MUSIQ (Ke et al., 2021)—and quantify dependence with three estimators that capture progressively more general relationships: the Pearson correlation (Lee Rodgers & Nicewander, 1988) for linear dependence, the Spearman rank correlation (Spearman, 1904) for monotonic dependence, and the Distance Correlation (Székely et al., 2007) for general nonlinear dependence. The resulting evidence substantiates our conclusion along two complementary fronts.

✦ **The correlation between CHR and perceptual metrics is dataset- and solver-dependent rather than intrinsic.** As the first and third rows of Table 2 show, the Pearson correlation between CHR and FID reverses sign across datasets—positive for SimObject ($r = 0.87$), negative for RealHand ($r = -0.97$), and positive again for PTI-Hand ($r = 0.20$)—and the Spearman correlation migrates likewise from positive to negative. The distance correlation, meanwhile, declines monotonically from 0.82 (SimObject) through 0.74 (RealHand) to 0.47 (PTI-Hand), indicating that increasingly complex generative content progressively erodes any association between CHR and FID; integrating

Table 2: Correlation of the counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR) with the perceptual metrics FID and MUSIQ. The PTI-Hand results are obtained by prompting a pretrained Stable-Diffusion-3.5-Medium model with ten hand-generation prompts to produce 5050 samples, matching the size of RealHand; FID for PTI-Hand is computed between the RealHand images and the model-generated samples. Here p is the p -value assessing the statistical significance of the correlation r or ρ , and $\dagger$ marks the DDPM results integrated with those from the various ODE solvers across sampling conditions. The key comparison is the contrast between the sign-reversing, magnitude-varying CHR columns and the uniformly strong, statistically significant NCFR columns: the perceptual metrics track non-counting failures but not counting hallucinations.

Metric	CHR vs. FID						NCFR vs. FID						CHR vs. MUSIQ						NCFR vs. MUSIQ					
	SimObject	RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	
Pearson (r)	0.87	0.01	-0.97	<0.01	0.20	0.69	0.99	<0.01	0.88	<0.01	0.93	<0.01	-0.02	0.95	<0.01	-0.97	<0.01	-0.99	<0.01					
Pearson † (r)	0.49	0.14	-0.53	0.13	—	—	0.96	<0.01	—	—	0.46	0.20	—	—	—	-0.98	<0.01	—	—					
Spearman (ρ)	0.84	<0.01	-0.97	<0.01	-0.14	0.78	0.99	<0.01	0.92	<0.01	0.80	0.01	0.42	0.33	-0.88	<0.01	-0.46	0.29						
Spearman † (ρ)	0.75	0.01	-0.51	0.15	—	—	0.95	<0.01	—	—	0.40	0.28	—	—	—	-0.91	<0.01	—	—					
Distance Correlation (r_d)	0.89	—	0.97	—	0.47	—	0.99	—	0.91	—	0.93	—	0.42	—	—	0.97	—	0.99	—					
Distance Correlation † (r_d)	0.82	—	0.74	—	—	—	0.95	—	—	—	0.71	—	—	—	—	0.97	—	—	—					

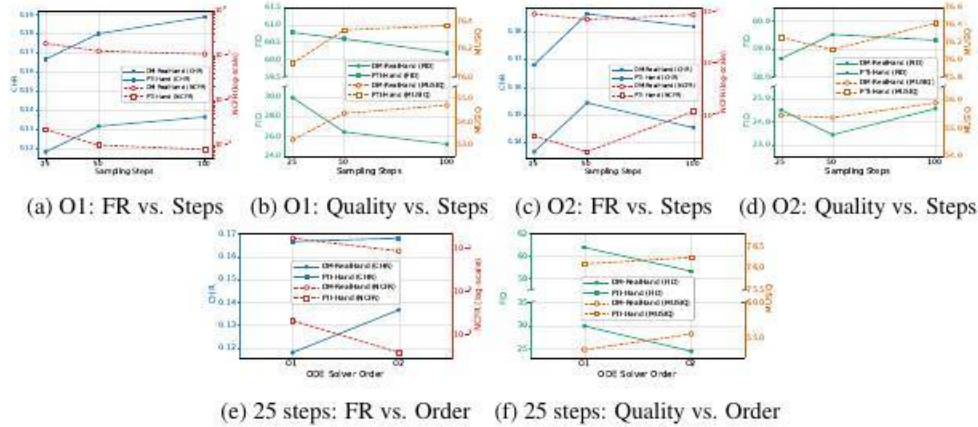


Figure 3: Trend consistency between DM-RealHand (ours) and PTI-Hand (pretrained Stable-Diffusion-3.5-Medium). O1 and O2 denote the first- and second-order ODE solvers (DPM-solver-1 / DPM-solver-2 for DM-RealHand; Euler / Heun for PTI-Hand); the failure-rate (FR) metrics are CHR and NCFR, and the quality metrics are FID and MUSIQ. Across variations in both sampling steps and solver order, the two models trace closely aligned trends, indicating that the qualitative behavior observed in our controlled RealHand setting recurs in a large-scale pretrained text-to-image model under a matched hand-generation task.

the DDPM results with those from the ODE solvers attenuates these correlations still further. Taken together, these patterns establish that FID carries no consistent signal about counting hallucinations.

✦ **The correlation between NCFR and perceptual metrics is consistently strong and positive.** In sharp contrast to the erratic CHR correlations, NCFR correlates strongly and robustly with the perceptual metrics in every setting: for NCFR versus FID, all coefficients exceed 0.88 at negligible p -values, denoting high statistical significance. The juxtaposition is decisive: the perceptual metrics track the perceptually salient non-counting failures faithfully while remaining blind to counting hallucinations—precisely the dissociation our study is designed to expose, and the empirical basis for our claim that perceptual quality and counting fidelity must be measured separately.

5.4 GENERALIZATION OF REALHAND FINDINGS TO PRETRAINED TEXT-TO-IMAGE MODELS

We next examine whether our findings extend beyond the controlled training regime to a large-scale pretrained text-to-image model. Applying the counting-ready indicator and counting model of Sec. 4.2.3, we evaluate Stable-Diffusion-3.5-Medium (Esser et al., 2024) directly on a matched hand-generation task (further details in Appendix A.8). As Figure 3 shows, the influence of solver

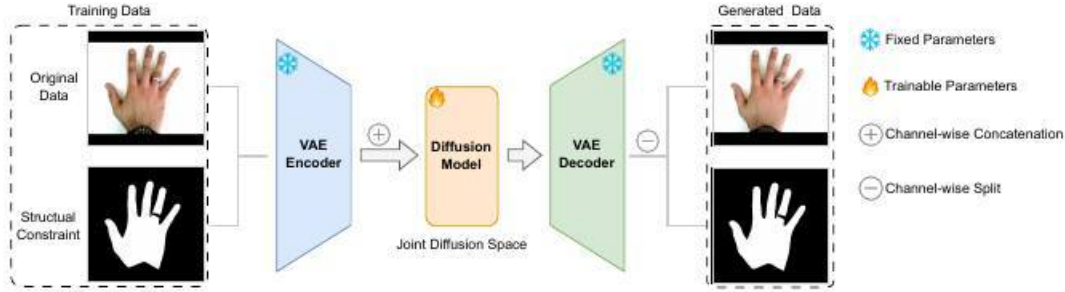


Figure 4: The proposed joint-diffusion model (JDM). JDM concatenates, channel-wise, the latent representations of the hand RGB image and its corresponding segmentation mask, expanding only the input and output channels of the denoising UNet. Denoising the two modalities jointly encourages anatomically plausible hand poses and correct finger topology during generation, and yields reduced CHR and NCFR on RealHand.

type and sampling steps on counting hallucinations in this pretrained model closely mirrors the trends established in our controlled RealHand experiments. This agreement provides encouraging evidence that the observations of Sec. 5.2 are not artifacts of our particular training setup, and that the proposed evaluation protocol is applicable beyond the datasets on which it was developed. The correlation results in Table 2 reinforce this reading, reproducing the same signature under the pretrained model: CHR correlates only weakly with the perceptual metrics while NCFR correlates strongly. We interpret these results as a first, targeted demonstration that our findings and protocol carry over to a modern latent text-to-image system on the hand-counting task; a comprehensive characterization across the full diversity of text-to-image models, prompts, and counting constraints remains an important direction for future work.

5.5 HOW TO REDUCE COUNTING HALLUCINATIONS?

The markedly lower counting hallucination rates observed for morphologically simpler objects (Observation 5, Sec. 5.2) indicate that diffusion models capture counting-based factuality far more reliably when the underlying visual structure is of low morphological complexity, as with masked objects. This regularity motivates a concrete hypothesis: if a diffusion model were to denoise general visual representations *jointly* with explicit, easily modeled structural constraints—object masks that encode basic topology and counts—within a shared latent space, such a joint learning paradigm ought to mitigate counting hallucinations. We refer to models that denoise general visual representations and structural constraints jointly as **joint-diffusion models** (JDM).

Concretely, we concatenate, channel-wise, the latent representation of the hand mask with that of the corresponding RGB image, as shown in Fig. 4, and fine-tune the UNet pretrained on CelebA-HQ (Karras et al., 2018)—following the RealHand configuration of Sec. 5.1—on the RealHand images together with their masks, the latter extracted with SAM-2 (Ravi et al., 2024). To accommodate both modalities, the input and output channels of the denoising UNet are expanded to match the combined latent dimensionality of the image and mask latents ($3 + 3 = 6$ channels), while all intermediate layers are left unchanged.

As Table 3 shows, JDM attains lower CHR and NCFR than the vanilla LDM under every sampling condition, evidence that supplying explicit structural constraints helps the model generate the structurally intricate objects most prone to counting ambiguity. The one cost is a higher FID—a degradation that follows directly from the added denoising burden of jointly modeling two modalities under a small-scale training regime, and that trades a measure of perceptual quality for improved structural realism, precisely the trade-off our correlation analysis shows FID cannot register. We present JDM as a preliminary, single-domain mitigation rather than a fully optimized method: it is demonstrated here only for hands and is not yet compared against alternative interventions such as rejection sampling with the counting detector, inference-time mask conditioning, or structural conditioning of the ControlNet family, which we regard as natural baselines for future study. Encouragingly, JDM further yields promising reductions in counting hallucinations when applied to large-scale pretrained diffusion models under carefully controlled finetuning, as detailed in Appendix A.9.

Table 3: Failure rates of the baseline LDM and our proposed JDM on the RealHand dataset across ODE solvers and sampling steps; lower is better. Although JDM jointly generates the hand RGB image and its segmentation mask, FID is computed on the RGB output alone so as to remain directly comparable with the baseline. JDM reduces CHR and NCFR under every configuration—the NCFR reduction being especially pronounced—at the cost of a higher FID, a metric our correlation analysis shows to be insensitive to the counting fidelity JDM improves.

Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fréchet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
Sampling Steps	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
LDM	12.95	13.85	14.55	14.48	15.99	15.43	18.06	12.51	10.63	9.33	7.22	8.94	29.86	26.43	25.21	24.48	23.42	24.54
JDM (ours)	11.41	10.67	10.51	10.12	9.66	10.07	2.94	2.42	2.29	3.51	4.63	2.26	37.69	35.51	34.68	34.63	32.07	33.89

6 CONCLUSION

This work takes a systematic step toward quantifying hallucinations in diffusion models and, in doing so, yields concrete insight into how factual generation can be improved. Centering on counting hallucinations, we introduce **CountHalluSet** together with a practical, reproducible methodology for measuring them. Through controlled quantification experiments, we establish that the numerical strategies commonly adopted to improve sampling fail to mitigate counting hallucinations when generating human hands, and our correlation analyses demonstrate that widely used perceptual metrics such as FID and MUSIQ do not consistently reflect their severity—a decoupling of perceptual quality from counting fidelity that sharpens with morphological complexity. Experiments on a large-scale pretrained text-to-image model, evaluated on a matched hand-generation task, corroborate the same qualitative behavior and indicate that our protocol transfers beyond its development datasets, and our joint-diffusion models chart a concrete, if still preliminary, route toward enforcing factual consistency in image generation. We hope that CountHalluSet and its evaluation protocol will provide a foundation for extending the systematic study of counting—and, ultimately, broader factual—hallucinations to richer conditional, compositional, and multi-object settings.

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A APPENDIX

A.1 POOF OF THE BOUND OF TOTAL ACCUMULATED DISTRIBUTIONAL ERROR IN ANCESTRAL SAMPLING

For the joint distributions of the ideal reverse process $q(\mathbf{x}_{T:0})$ and the model reverse process $p_\theta(\mathbf{x}_{T:0})$, the KL divergence is given by

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})) = \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_{T:0})}{p_\theta(\mathbf{x}_{T:0})} \right]. \quad (\text{A.1})$$

By the Markov property, both reverse processes $q(\mathbf{x}_{T:0})$ and $p_\theta(\mathbf{x}_{T:0})$ factorize as

$$q(\mathbf{x}_{T:0}) = q(\mathbf{x}_T) \prod_{t=1}^T q(\mathbf{x}_{t-1} | \mathbf{x}_t), \quad (\text{A.2})$$

$$p_\theta(\mathbf{x}_{T:0}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t). \quad (\text{A.3})$$

Applying the chain rule to decompose the KL divergence accordingly yields

$$\begin{aligned} \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})) &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T) \prod_{s=1}^T q(\mathbf{x}_{s-1} | \mathbf{x}_s)}{p_\theta(\mathbf{x}_T) \prod_{s=1}^T p_\theta(\mathbf{x}_{s-1} | \mathbf{x}_s)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} + \sum_{t=1}^T \log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_T)} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} \right] + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} \left[\mathbb{E}_{q(\mathbf{x}_{t-1} | \mathbf{x}_t)} \left[\log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \middle| \mathbf{x}_t \right] \right] \\ &= \mathcal{D}_{\text{KL}}(q(\mathbf{x}_T)||p_\theta(\mathbf{x}_T)) + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t)||p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]. \quad (\text{A.4}) \end{aligned}$$

By the Data Processing Inequality (Cover & Thomas, 2012), marginalizing out—equivalently, discarding—random variables can never increase the KL divergence:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0)) \leq \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})). \quad (\text{A.5})$$

Combining Eq. (A.4) with Eq. (A.5) then yields **the bound on the total accumulated distributional error**:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0)) \leq \underbrace{\mathcal{D}_{\text{KL}}(q(\mathbf{x}_T)||p_\theta(\mathbf{x}_T))}_{\text{diffused-prior gap}} + \underbrace{\sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t)||p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]}_{\text{accumulated transition errors caused by the model prediction error}}, \quad (\text{A.6})$$

where equality holds only in the ideal case $q(\mathbf{x}_T) = p_\theta(\mathbf{x}_T)$ and $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = q(\mathbf{x}_{t-1} | \mathbf{x}_t)$ for all t and $\mathbf{x}_t$, at which point the diffusion model recovers the underlying data distribution $q(\mathbf{x}_0)$ exactly. This upper bound on the overall KL divergence $\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0))$ between the true and the model-generated data coincides, once the data entropy $H(\mathbf{x}_0)$ is excluded, with the variational bound derived in DDPM (Ho et al., 2020); it follows that optimizing DDPM’s variational loss directly tightens this explicit upper bound on the discrepancy between true and generated data.

Diffused-prior gap. The first error source is the mismatch between the distribution of the ground-truth diffused data $q_T(\mathbf{x}_T) = q_{0:T}(\mathbf{x}_T|\mathbf{x}_0)$ and the prior distribution $p_\theta(\mathbf{x}_T)$ used for sampling—a mismatch that arises precisely because the number of diffusion steps T is finite in practice. This discrepancy is quantified by a general distributional metric:

$$\mathcal{D}_{\text{KL}}(q_T(\mathbf{x}_T)||p_\theta(\mathbf{x}_T)), \quad p_\theta(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (\text{A.7})$$

where $\mathcal{D}_{\text{KL}}$ denotes the Kullback–Leibler (KL) divergence; alternative distributional metrics such as FID or the Wasserstein distance may equally serve to evaluate this discrepancy. Present from the

outset, the diffused-prior gap induces a deviation that propagates through the entire denoising process and manifests as a sustained output error.

Noise model prediction error. The second error source is the noise prediction model itself. Its estimate $\epsilon_\theta(\mathbf{x}_t, t)$ only approximates the true noise ϵ , an approximation feasible in principle by the *Universal Approximation Theorem* (Hornik et al., 1989; Nielsen, 2015; Yarotsky, 2022). Writing the prediction error as $\Delta\epsilon_t(\mathbf{x}_t) = \epsilon - \epsilon_\theta(\mathbf{x}_t, t)$, Eq. (5) implies that any such discrepancy induces a corresponding deviation in the predicted mean at step t :

$$\Delta\mu_t(\mathbf{x}_t) = -\frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\bar{\alpha}_t}} \Delta\epsilon_t(\mathbf{x}_t). \quad (\text{A.8})$$

Because each reverse step is conditioned on the mean from the preceding iteration, this bias accumulates along the denoising trajectory and thereby compounds into the overall prediction error. Notably, this decomposition is explanatory rather than predictive: it identifies the two mechanisms by which sampling error accrues, but we do not claim it quantitatively forecasts the counting hallucination rates reported in Sec. 5.2.

A.2 POOF OF THE TOTAL ACCUMULATED TRAJECTORY ERROR IN ODE-BASED SAMPLING

Starting from Eq. (7), we analyze a one-step numerical solver as a representative case. Let $\mathbf{x}^*(\tau)$ denote the ideal trajectory satisfying this ODE under the true score function $h^*(\mathbf{x}(\tau), \tau)$ and initialized from a ground-truth condition $\mathbf{x}^*(T)$ drawn from the true diffused distribution $q_T(\mathbf{x}_T)$. Correspondingly, let $\hat{\mathbf{x}}_k$ denote the trajectory produced by a numerical ODE solver at the discrete time steps $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, and let $\hat{h}$ be the estimated model function. Our objective is to quantify the total accumulated trajectory error of a given numerical trajectory $\hat{\mathbf{x}}_0, \dots, \hat{\mathbf{x}}_N$. Defining the per-step trajectory error as $e_k = \hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)$, we consider the transition from τ_k to $\tau_{k+1} = \tau_k - \Delta t$.

Ideal path evolution. The variation-of-constants formula supplies the exact solution over one step, from τ_k down to τ_{k+1} :

$$\mathbf{x}^*(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}^*(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h^*(\mathbf{x}^*(u), u) du, \quad (\text{A.9})$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state transition operator for the linear part $\frac{dz}{d\tau} = f(\tau)z$, and the integration limits run backward in time.

Numerical path evolution. A one-step numerical solver approximates this solution, and its step departs from the exact evolution governed by $\hat{h}$ starting from $\hat{\mathbf{x}}_k$. Let $\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} \mid \hat{\mathbf{x}}_k, \hat{h})$ denote the exact solution of the ODE under $\hat{h}$ at τ_{k+1} , initialized from $\hat{\mathbf{x}}_k$ at τ_k :

$$\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} \mid \hat{\mathbf{x}}_k, \hat{h}) = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u \mid \hat{\mathbf{x}}_k, \hat{h}), u) du. \quad (\text{A.10})$$

The numerical method thereby introduces a local truncation error δ_k :

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} \mid \hat{\mathbf{x}}_k, \hat{h}) + \delta_k. \quad (\text{A.11})$$

so that

$$\hat{\mathbf{x}}_{k+1} = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u \mid \hat{\mathbf{x}}_k, \hat{h}), u) du + \delta_k. \quad (\text{A.12})$$

Error recurrence. Subtracting the ideal evolution Eq. (A.9) from the numerical evolution Eq. (A.12) yields the one-step error recurrence:

$$\begin{aligned} e_{k+1} &= \hat{\mathbf{x}}_{k+1} - \mathbf{x}^*(\tau_{k+1}) \\ &= \mathcal{G}(\tau_{k+1}, \tau_k) (\hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u \mid \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k. \end{aligned} \quad (\text{A.13})$$

Accumulated error formulation. Unrolling this recurrence from $k = 0$ to $N - 1$, and writing $\mathcal{G}(\tau_j, \tau_k) = \mathcal{G}(\tau_j, \tau_{j-1}), \dots, \mathcal{G}(\tau_{k+1}, \tau_k)$ for $j > k$ with $\mathcal{G}(\tau_k, \tau_k) = \mathbf{I}$, repeated substitution yields

$$e_N = \mathcal{G}(\tau_N, \tau_0)e_0 + \sum_{k=0}^{N-1} \mathcal{G}(\tau_N, \tau_{k+1}) \cdot \left(\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k \right). \quad (\text{A.14})$$

Substituting $\tau_N = 0$, $\tau_0 = T$, $e_N = \hat{\mathbf{x}}_N - \mathbf{x}^*(0)$, and $e_0 = \hat{\mathbf{x}}_0 - \mathbf{x}^*(T)$ then delivers

$$\hat{\mathbf{x}}_N - \mathbf{x}^*(0) = \underbrace{\mathcal{G}(0, T)(\hat{\mathbf{x}}_0 - \mathbf{x}^*(T))}_{\text{propagated initial error}} + \sum_{k=0}^{N-1} \mathcal{G}(0, \tau_{k+1}) \cdot \left(\underbrace{\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) - h^*(\mathbf{x}^*(u), u) \right] du}_{\text{single step model prediction error contribution}} + \underbrace{\delta_k}_{\text{single step truncation error}} \right). \quad (\text{A.15})$$

where the propagated-initial-error term captures the deterministic contribution of a specific initial-error instance to the final path error, its statistical origin residing in the distributional diffused-prior gap. As in the ancestral-sampling analysis, this decomposition is offered as a structural account of the sources of trajectory error—the propagated initial error, the per-step model prediction error, and the truncation error controlled by step size and solver order—rather than as a quantitative predictor of the empirical hallucination rates.

A.3 THE DISCREPANCY BETWEEN DDPM AND DDIM IN THE FORWARD AND REVERSE PROCESS

As stated in Eq. (2), the forward process of DDPM (Ho et al., 2020) is

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (\text{A.16})$$

and its reverse process is defined as

$$\mathbf{x}_{t-1}^{\text{DDPM}} = \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \frac{1 - \bar{\alpha}_t}{\sqrt{\bar{\alpha}_t} \sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \epsilon. \quad (\text{A.17})$$

For DDIM (Song et al., 2021), the forward process is identical to Eq. (A.16), whereas its deterministic reverse process (with $\eta = 0$) is defined as

$$\begin{aligned} \mathbf{x}_{t-1}^{\text{DDIM}} &= \sqrt{\frac{\bar{\alpha}_{t-1}}{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(\mathbf{x}_t, t)) + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon_\theta(\mathbf{x}_t, t) \\ &= \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \left(\frac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{\bar{\alpha}_t}} - \sqrt{1 - \bar{\alpha}_{t-1}} \right) \epsilon_\theta(\mathbf{x}_t, t). \end{aligned} \quad (\text{A.18})$$

Although DDPM and DDIM share an identical forward (diffusion) process, DDIM is not recovered by merely deleting the stochastic term from DDPM’s reverse (denoising) process: the two methods employ genuinely different coefficient formulations for the predicted-noise term, as the contrast between the two reverse-process expressions above makes plain.

A.4 DATASET EXAMPLES AND DETAILS

Figure A1 presents representative examples from the three datasets used in this study. Table A1 shows that each dataset is balanced across its categories, and Table A2 further establishes that the distribution of images over distinct object counts is comparable across datasets, confirming the absence of object-count bias that might otherwise confound the measured hallucination rates. The datasets appear in the same order in both tables, and the category order in Table A1 is as follows:

- **ToyShape:** triangle, square, pentagon;
- **SimObject:** mug, apple, clock;
- **RealHand:** finger.

Table A1: Number of objects per category for each dataset, evidencing the balanced category distribution within every dataset.

	ToyShape	SimObject	RealHand
Category 1	20,027	19,916	25,250
Category 2	19,902	20,101	-
Category 3	19,978	19,974	-

Table A2: Number of images grouped by object count for each dataset, illustrating the comparable count distribution across datasets and the resulting absence of object-count bias.

	ToyShape	SimObject	RealHand
1 object	10,003	10,022	-
2 objects	10,087	9,965	-
3 objects	9,910	10,013	-
5 objects	-	-	25,250
Total	30,000	30,000	25,250

A.5 EXAMPLES OF COUNTING HALLUCINATIONS

We present representative counting hallucinations for all three datasets in Fig. A2. For ToyShape and SimObject, the counting hallucinations either contain two or more instances of at least one category, or contain no instances at all while retaining a normal background, as shown in Fig. A2(a) and Fig. A2(b). For RealHand, counting-hallucinated samples are distinguished by incorrect finger counts (e.g., 4 or 6 fingers), as shown in Fig. A2(c). The last two images in the first row of Fig. A2(c) are especially instructive: they appear at first glance to display five fingers, yet one finger is partially incomplete and visibly artifacted—leaving four valid fingers—so that these samples are correctly classified as counting hallucinations rather than as counting-valid.

A.6 EXAMPLES OF YOLO PREDICTIONS OF FINGERTIPS

Figure A3 presents representative YOLO predictions on the RealHand dataset. The YOLO detector localizes fingertips accurately across diverse hand poses and orientations, spanning both dorsal and palmar views. It can nonetheless be defeated by anatomically implausible configurations, such as hands with duplicated thumbs (e.g., the penultimate sub-figure of the first row and the last sub-figure of the last row in Fig. A3(a)) or hands with five middle fingers but no thumb (e.g., the last sub-figure of the first row and the penultimate sub-figure of the last row in Fig. A3(a)). Such counting-valid samples are still detected as five-finger hands and thus remain indistinguishable, by finger count alone, from anatomically valid ones—a limitation inherent to a count-based criterion.

Figure A3(b) shows counting-hallucinated samples for which YOLO predicts incorrect fingertip counts (e.g., 3, 4, 6, or 7 fingertips), while Figure A3(c) illustrates rare edge cases (<1%) in which the detector fails on fingertips with unclear boundaries, cracked fingers, or floating artifacts. These edge cases can be partially mitigated by tuning YOLO’s inference hyperparameters (e.g., the IoU and confidence thresholds), but doing so degrades detection on the large majority of counting-valid samples—a genuine trade-off that motivates our choice of a single fixed configuration. Because that one detection configuration is applied uniformly across all RealHand evaluations, these residual cases bias every configuration equally and therefore leave the comparative conclusions of Sec. 5.2 intact, even though eliminating them entirely would be a challenging and low-reward undertaking.

A.7 EXAMPLES OF THREE CLASSES OF GENERATED HAND IMAGES

Figure A4 shows representative examples of the three classes of generated hand images: counting-valid, counting-hallucinated, and non-counting failure samples. As illustrated in Fig. 2, the Real-Hand evaluation procedure first applies the counting-ready indicator to separate the counting-ready samples—the counting-valid and counting-hallucinated cases—from images unsuitable for counting,

the latter typically exhibiting severe visual artifacts such as distorted, cracked, partially visible, or floating fingers. The counting-ready samples are then adjudicated as counting-valid or counting-hallucinated according to whether the number of detected valid fingertips equals five, completing the two-stage protocol.

A.8 EXTENDED DETAILS OF PTI-HAND EXPERIMENTS

For the PTI-Hand experiments, we prompt Stable-Diffusion-3.5-Medium with the 10 distinct hand-generation prompts listed below.

1. A single human hand, front view, all fingers in frame and visible.
2. A complete human hand, seen from the front, showing all fingers clearly, minimal background.
3. An isolated single hand, viewed from directly above, showing all fingers, simple studio background.
4. Overhead shot of a person’s hand, entire hand in frame, all fingers visible, plain background.
5. One hand, top view, five fingers visible, simple background.
6. A single hand, front or back view, flat against a surface, all fingers in frame and fully visible.
7. A single human hand, top-down view, all fingers visible, resting on a plain white surface.
8. A single human hand seen from above, fingers spread naturally, minimal background.
9. A human hand, top view, palm facing downward, all fingers clearly in frame, neutral background.
10. A single human hand captured from overhead, fingers extended, plain background with soft lighting.

Our definition of hallucination presupposes that the model generates patterns never observed during training—an assumption difficult to verify for large-scale pretrained diffusion models, whose full training distribution is inaccessible. To keep the quantification of counting hallucinations feasible in this regime, we therefore adopt a deliberately conservative criterion: any generated hand image containing six or more fingers is treated as a counting-hallucinated sample. This criterion is well justified by an asymmetry in the real-image distribution. Whereas real images may legitimately depict fewer than five visible fingers owing to occlusion, articulation, or viewpoint, real data are exceedingly unlikely to contain hands with six or more fingers. A finger count of six or more thus constitutes a reliable and empirically verifiable signal of counting hallucination, even if it necessarily forgoes the under-counting cases that the closed RealHand setting can additionally capture. Generated examples appear in Fig. A5.

A.9 EXTENDED DETAILS, RESULTS, AND DISCUSSION OF THE JOINT DIFFUSION MODEL

A.9.1 IMPLEMENTATION DETAILS IN TABLE 3

For the JDM experiments reported in Table 3, we fine-tune the denoising UNet for 120k steps, as against 80k for the vanilla LDM. This larger training budget is necessary because extending the latent space from three to six channels introduces a substantial mismatch with the pretrained latent distribution of the LDM; the additional optimization is required to close this distributional gap and thereby allow JDM’s modeling capacity to materialize fully.

A.9.2 FINE-TUNING PRETRAINED TEXT-TO-IMAGE MODELS WITH JOINT-DIFFUSION TRAINING

Importantly, JDM is to be understood as a **training paradigm** for improving factual generation in diffusion models, not as a superior inference strategy. On one hand, because JDM extends only the input and output channels of the UNet while leaving every intermediate-layer dimension unchanged, it **introduces negligible inference-time overhead relative to the vanilla LDM**. On the other hand,

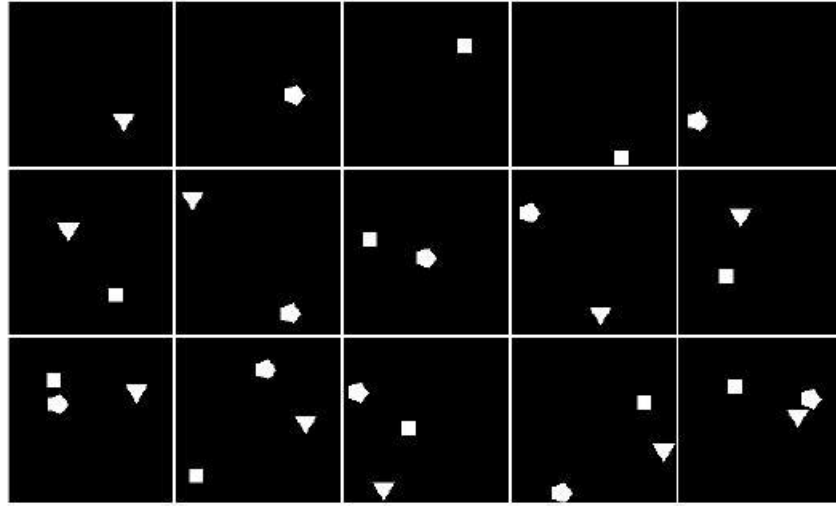
Table A3: Failure rates and image quality for finetuning SD-1.5 versus finetuning SD-1.5 with the joint-diffusion training paradigm on RealHand. The first row (baseline) reports finetuning Stable-Diffusion-1.5 on RealHand; the second row reports the joint-diffusion variant, obtained by freezing all intermediate layers of the baseline and finetuning only the input and output layers with extended channel dimensions. Lower values are better. Even under this deliberately constrained setup, the joint-diffusion paradigm lowers CHR at 25 and 50 steps under both solvers, at the expected cost of higher FID.

Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fr�chet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
SD-1.5	0.3362	0.3404	0.3630	0.3396	0.3376	0.3362	0.0059	0.0032	0.0020	0.0050	0.0026	0.0018	61.51	57.71	57.70	58.53	57.57	56.69
SD-1.5 with joint-diffusion finetuning	0.2838	0.3222	0.4145	0.3053	0.3277	0.3558	0.0947	0.0511	0.0487	0.0549	0.0362	0.0327	70.65	79.22	102.39	70.80	72.84	83.07

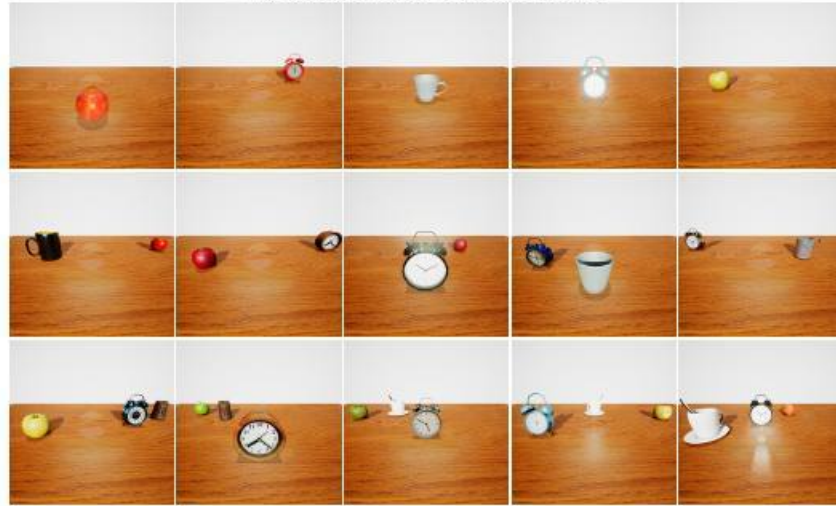
augmenting a pretrained text-to-image model with additional mask channels **perturbs the manifold geometry of the pretrained latent space and disrupts the original denoising distribution**, which renders direct finetuning of such a model nontrivial.

This challenge notwithstanding, we still obtain promising reductions in counting hallucinations when applying JDM to large-scale pretrained diffusion models under carefully controlled finetuning. In practice, we find that aligning JDM’s extended latent space with the highly rigid latent geometry of large pretrained models is difficult at the limited scale of RealHand. To strike a workable balance, we therefore adopt a two-stage finetuning strategy: we first finetune a Stable-Diffusion-1.5 model on RealHand, then freeze all intermediate UNet layers and finetune only the input and output layers with the extended channels. As Table A3 shows, even under this deliberately constrained setup the joint-diffusion paradigm consistently lowers the counting hallucination rate at 25 and 50 sampling steps across both solvers.

Taken together, the evidence assembled here identifies JDM as a promising direction for enforcing factual consistency in image generation. We are careful, however, not to overstate it: the present results are confined to hands and to a small-scale regime, they trade FID for counting fidelity, and they are not yet benchmarked against the mitigation baselines enumerated in Sec. 5.5. We accordingly regard **a systematic study of JDM’s behavior under genuinely large-scale training, alongside a controlled comparison to these baselines, as an important and compelling avenue for future work.**



(a) Examples in the ToyShape dataset

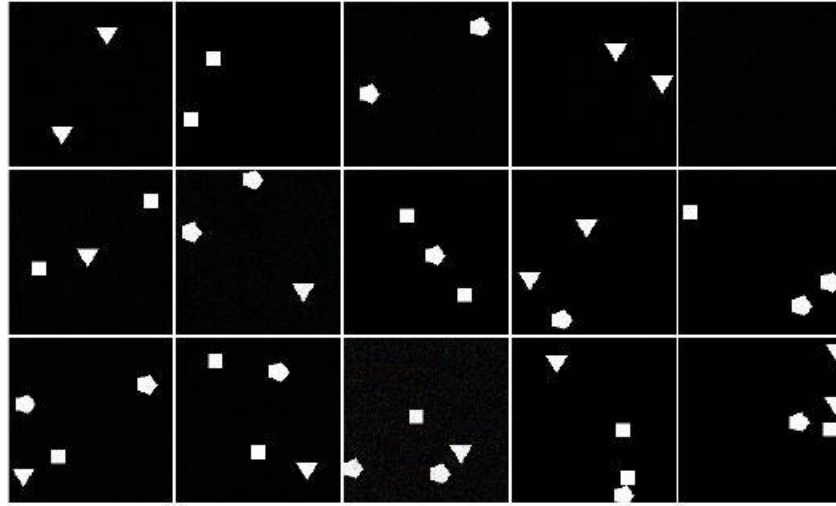


(b) Examples in the SimObject dataset



(c) Examples in the RealHand dataset

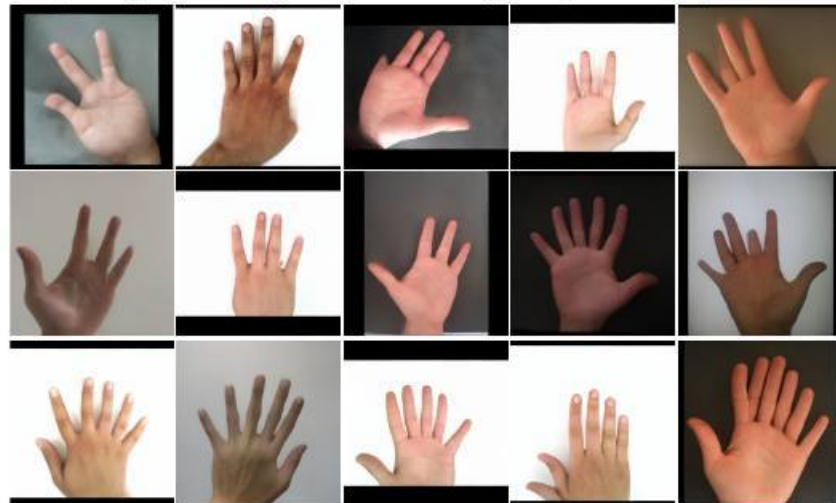
Figure A1: Representative examples from the three datasets of the CountHalluSet suite: ToyShape, SimObject, and RealHand.



(a) Examples of counting hallucinations corresponding to the ToyShape dataset

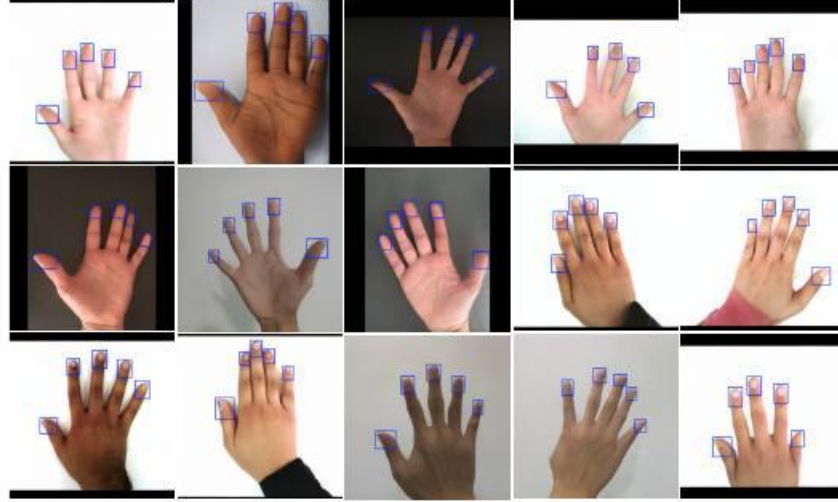


(b) Examples of counting hallucinations corresponding to the SimObject dataset

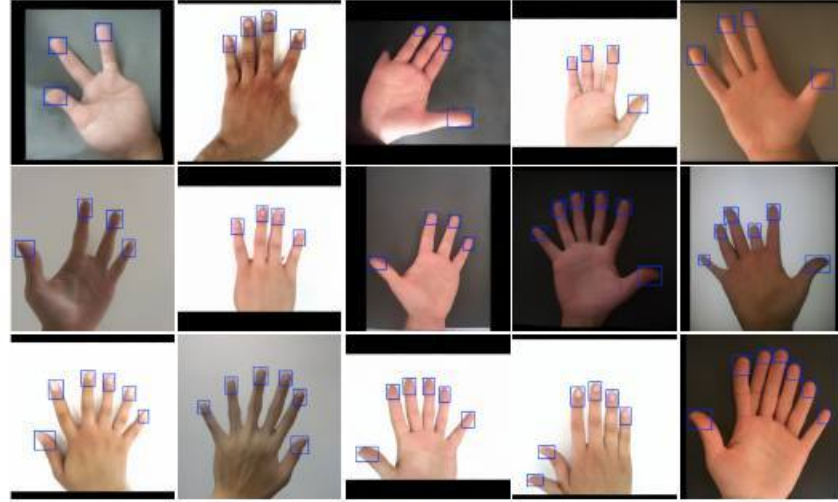


(c) Examples of counting hallucinations corresponding to the RealHand dataset

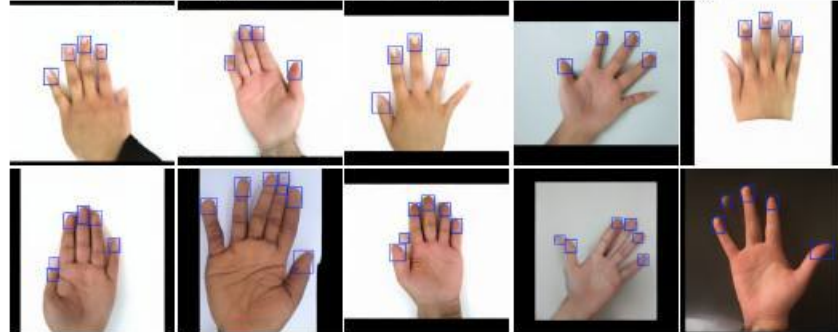
Figure A2: Representative counting hallucinations for the ToyShape, SimObject, and RealHand datasets.



(a) Counting-valid YOLO predictions with five fingers

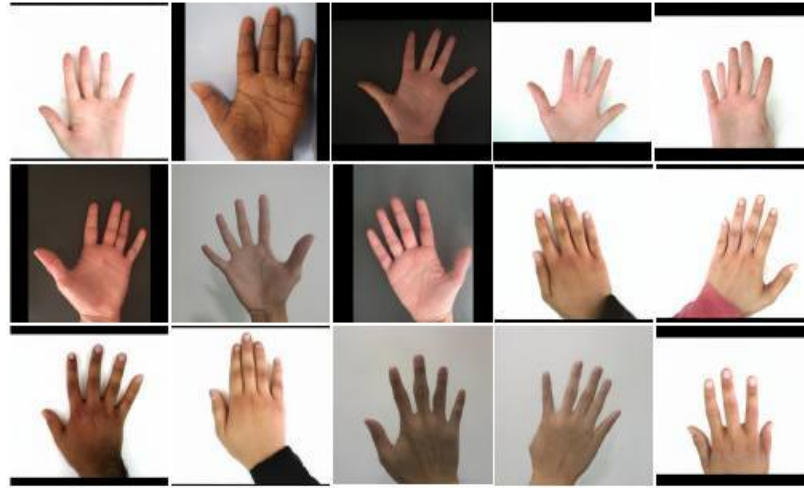


(b) Counting-hallucinated YOLO predictions with not five fingers

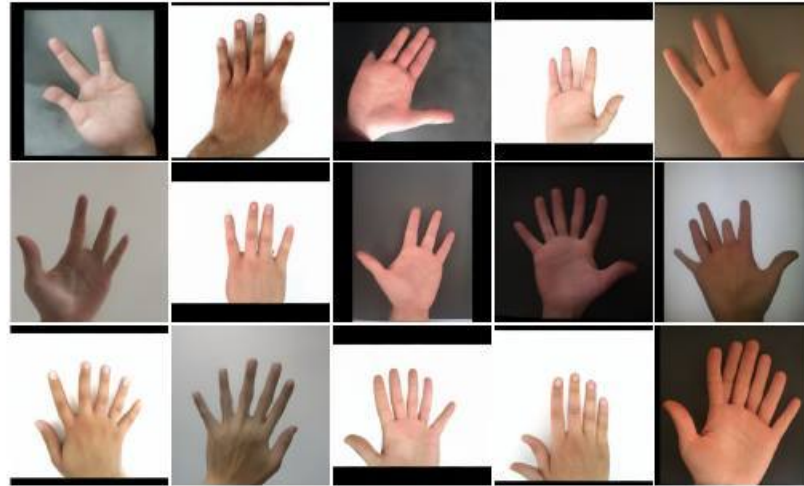


(c) Rare edge cases where YOLO predictions fail

Figure A3: YOLO predictions on generated RealHand images. (a) Counting-valid samples for which YOLO correctly predicts five fingertip bounding boxes, including anatomically unusual cases with two thumbs. (b) Counting-hallucinated samples with incorrect fingertip counts (e.g., 3, 4, 6, or 7 detected fingertips). (c) Rare edge cases that the counting-ready indicator fails to filter out and that YOLO in turn struggles to resolve, such as blurred, incomplete, or anatomically disconnected fingers.



(a) Counting-valid samples (five valid fingers)



(b) Counting-hallucinated samples (not five valid fingers)



(c) Non-counting failure samples (unsuitable for counting)

Figure A4: The three classes of generated hand images produced under the controlled RealHand setting. (a) Counting-valid: images with five clearly discernible fingers. (b) Counting-hallucinated: images with an incorrect number of valid fingers (e.g., 4 or 6). (c) Non-counting failures: images unsuitable for counting owing to severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers.



(a) Counting-valid samples (five valid fingers)



(b) Counting-hallucinated samples (more than 5 fingers)



(c) Non-counting failure samples (unsuitable for counting)

Figure A5: The three classes of generated hand images produced by pretrained Stable-Diffusion-3.5-Medium (PTI-Hand). (a) Counting-valid: images with five clearly discernible fingers. (b) Counting-hallucinated: images with more than five valid fingers (e.g., 6 or 7), in accordance with the conservative six-or-more criterion adopted for the pretrained regime. (c) Non-counting failures: images unsuitable for counting owing to severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers.